

Sequences, Series, and the Binomial Theorem

Answer Key

Algebra 2

Quick Answers

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|-------------------------------------|------------------------------------|
| 1. 15 | 7. 1215 |
| 2. 56 | 8. 243 |
| 3. $x^3 + 6x^2 + 12x + 8$ | 9. -9, 10 |
| 4. 32 | 10. 80 |
| 5. 40 | 11. $a^3 + 9a^2b + 27ab^2 + 27b^3$ |
| 6. $16x^4 - 32x^3 + 24x^2 - 8x + 1$ | 12. 1 |
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Common wrong answers

2. *If they got 336: forgot the $k!$ in the denominator (permutations, not combinations)*
5. *If they got 10: used the binomial coefficient alone and forgot the 2 raised to a power*
7. *If they got 135: raised 3 to the wrong power (used the y -exponent instead of $6 - k$)*
10. *If they got 10: left the 2 inside $2x$ unraised, using only the binomial coefficient*
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1. Evaluate $\binom{6}{2}$.

Step 1 (write the formula): $\binom{6}{2} = \frac{6!}{2!4!}$. Step 2 (cancel before multiplying): $\frac{6 \cdot 5 \cdot 4!}{2 \cdot 1 \cdot 4!} = \frac{30}{2}$.

Step 3: $\boxed{15}$ — also read off row 6 of Pascal's triangle, 1, 6, 15, 20, 15, 6, 1.

2. Evaluate $\binom{8}{3}$ and say what it counts.

Step 1: $\binom{8}{3} = \frac{8!}{3!5!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1}$. Step 2: $\frac{336}{6} = \boxed{56}$. Step 3 (meaning): it counts the ways to choose which 3 of the 8 factors of $(u + v)^8$ contribute a v , which is exactly the coefficient of u^5v^3 .

3. Expand $(x + 2)^3$.

Step 1 (write the general term): $\binom{3}{k}x^{3-k}2^k$ for $k = 0, 1, 2, 3$, with coefficients 1, 3, 3, 1 from row 3. Step 2 (build the terms): $1 \cdot x^3$, $3 \cdot x^2 \cdot 2$, $3 \cdot x \cdot 4$, $1 \cdot 8$. Step 3 (collect): $\boxed{x^3 + 6x^2 + 12x + 8}$.

4. Add the six coefficients in row 5.

Step 1 (list the row): $\binom{5}{0}, \dots, \binom{5}{5}$ are 1, 5, 10, 10, 5, 1. Step 2 (add): $1+5+10+10+5+1 = \boxed{32}$. Step 3 (structure): every row of Pascal's triangle sums to 2^n , and $2^5 = 32$ — setting $u = v = 1$ in $(u + v)^5$ says the same thing.

5. Coefficient of x^3 in $(x + 2)^5$.

Step 1 (pick the right term): the general term is $\binom{5}{k}x^k2^{5-k}$, so x^3 needs $k = 3$ — no other term contributes. Step 2 (evaluate the two pieces): $\binom{5}{3} = 10$ and $2^{5-3} = 4$. Step 3 (multiply): $10 \cdot 4 = \boxed{40}$. The coefficient is not 10 alone; the power of 2 belongs to it.

6. Expand $(2x - 1)^4$.

Step 1 (general term): $\binom{4}{k}(2x)^{4-k}(-1)^k$, with row 4 coefficients 1, 4, 6, 4, 1. Step 2 (raise the whole first term, keep the signs): $16x^4$, $4 \cdot 8x^3 \cdot (-1) = -32x^3$, $6 \cdot 4x^2 = 24x^2$, $4 \cdot 2x \cdot (-1) = -8x$, $+1$. Step 3 (collect): $\boxed{16x^4 - 32x^3 + 24x^2 - 8x + 1}$.

7. Coefficient of y^2 in $(3 + y)^6$.

Step 1 (match the pattern): with the constant written first, the general term is $\binom{6}{k}3^{6-k}y^k$, so y^2 means $k = 2$. Step 2 (evaluate): $\binom{6}{2} = 15$ and $3^{6-2} = 3^4 = 81$. Step 3 (multiply): $15 \cdot 81 = \boxed{1215}$. Using 3^2 instead of 3^4 is the usual slip and gives 135.

8. Sum of all coefficients of $(2x + 1)^5$.

Step 1 (write the sum): the coefficients are $\binom{5}{n}2^n$, so the total is $\sum_{n=0}^5 \binom{5}{n}2^n$. Step 2 (add term by term): $1 + 10 + 40 + 80 + 80 + 32 = 243$. Step 3 (confirm structurally): setting $x = 1$ gives $(2 + 1)^5 = 3^5 = \boxed{243}$ — the same number, which is the fast check.

9. For which n is $\binom{n}{2} = 45$?

Step 1 (turn the coefficient into a polynomial): $\binom{n}{2} = \frac{n(n-1)}{2}$, so $\frac{n(n-1)}{2} = 45$. Step 2 (solve): $n^2 - n - 90 = 0$, so $(n - 10)(n + 9) = 0$. Step 3 (choose the root that fits): an exponent must be a positive integer, so $\boxed{n = 10 \quad (n = -9 \text{ rejected})}$.

10. Coefficient of x^3y^2 in $(2x + y)^5$.

Step 1 (locate the term): the general term is $\binom{5}{k}(2x)^ky^{5-k}$, and x^3y^2 forces $k = 3$. Step 2 (evaluate both pieces): $\binom{5}{3} = 10$ and $(2)^3 = 8$ from the $2x$. Step 3 (multiply): $10 \cdot 8 = \boxed{80}$, so the term is $80x^3y^2$.

11. Expand $(a + 3b)^3$.

Step 1 (general term): $\binom{3}{k}a^{3-k}(3b)^k$ with row 3 coefficients 1, 3, 3, 1. Step 2 (raise $3b$ each time): a^3 ; $3 \cdot a^2 \cdot 3b = 9a^2b$; $3 \cdot a \cdot 9b^2 = 27ab^2$; $27b^3$. Step 3 (collect): $\boxed{a^3 + 9a^2b + 27ab^2 + 27b^3}$. The 1, 3, 3, 1 of row 3 is still visible, multiplied by the powers of 3.

12. Synthesis: $(3x - 2)^4$, **general term and sum of coefficients.**

Step 1 (general term): $\binom{4}{n}(3)^{4-n}(-2)^n x^{4-n}$, so the coefficient list is $\binom{4}{n}3^{4-n}(-2)^n$ for $n = 0$ to 4. Step 2 (add the five coefficients): $81 - 216 + 216 - 96 + 16 = \boxed{1}$. Step 3 (why it had to be 1): substituting $x = 1$ makes every power of x equal to 1, so the sum of the coefficients is just $(3 - 2)^4 = 1$. The large positive and negative coefficients must cancel down to that.